HERIOT-WATT UNIVERSITY

SCHOOL OF MATHEMATICAL & COMPUTER SCIENCES

DEPARTMENT OF ACTUARIAL MATHEMATICS & STATISTICS

F78AA1
ACTUARIAL AND FINANCIAL MATHEMATICS A

December 2009
SOLUTIONS

Total marks: 60

Only approved calculators may be used.

1.

(i) (a) ${}_5|\ddot{a}_{\overline{20}|} = v^4(\frac{1-v^{20}}{i}) = 7.08941$ where $v = 1.04^{-2}$

(b) $s^{(4)}_{\overline{15}|} = \frac{1}{4}\frac{(1.04^{\frac{1}{2}})^{60}-1}{1.04^{\frac{1}{2}}-1} = 28.3201$

[4 marks]

(ii) $a_{\overline{15}|} = v + v^2 + \cdots + v^{15}$
$= v^{15}[(1+i)^{14} + (1+i)^{13} + \cdots + 1] = v^{15}s_{\overline{15}|}$

[3 marks]

[**Total 7 marks**]

2. (i) Using $v = \frac{1}{1.10}$

(a) $300a_{\overline{20}|} = 300\frac{(1-v^{20})}{i} = £2,554.07$

(b) $300(v + 1.12v^2 + \cdots + 1.12^{19}v^{20}) = \frac{300}{1.12}(\frac{w^{21}-w}{w-1}) = £6,507.90$
where $w = \frac{1.12}{1.10}$

(c) $300(v + v^2 + \cdots + v^{20}) + 36(v^2 + 2v^3 + \cdots + 19v^{20})$
$= 300a_{\overline{20}|} + 36v(Ia)_{\overline{19}|} = £2,554.07 + £1,994.65 = £4,548.72$
where $(Ia)_{\overline{19}|} = \frac{\ddot{a}_{\overline{19}|}-19(1.1)^{-19}}{0.1}$

[9 marks]

(ii) compounding increases gives increases on increases, so this means that all payments are higher after the second payment.

[2 marks]

(iii) $d = 6\% \Rightarrow i = \frac{1}{1-d} - 1 = 6.38298\%$
$3\delta = 0.21$
$1,000(1.0638298)^2e^{0.21} = £1,396.20$

[4 marks]

[**Total 15 marks**]

3. (a)

$$\frac{D(1-t_1)}{P} \leq i \leq \frac{1}{P}(D(1-t_1) + \frac{(R-P)}{n}(1-t_2))$$

$$\Rightarrow \frac{.05 \times 0.8}{1.1} \leq i \leq \frac{1}{1.1}(.05 \times 0.8 + \frac{0.1}{12}(0.6))$$

$$\Rightarrow 3.636\% \leq i \leq 4.091\%$$

Let $f(i) = 22,000 - 800a_{\overline{12}|} - (24,000 - (2,000 \times 0.4))v^{12} = 22,000 - 800a_{\overline{12}|} - 23,200v^{12}$
$f(4.091\%) = 191.88$
$f(3.636\%) = -782.49$
$\Rightarrow i \approx 4.001\%$ using linear interpolation
Exact solution is 3.999%

[6 marks]

(b) $5 \times \frac{213.4}{192.2} \times 200 = £1,110.30$

[2 marks]

(c) $20,000 \times \frac{213.4}{192.2} \times 1.05^{11} = £37,979.86$

[3 marks]

(d)

$$P = 5v^{\frac{1}{2}}\frac{213.4}{192.2}(1 + 1.05v + \cdots + 1.05^{11}v^{11}) + 100v^{11.5}1.05^{11}\frac{213.4}{192.2}$$

$$\Rightarrow P = 49.8153 + 63.461 = 113.276$$

per £100 nominal

So $P = 113.276 \times 200 = £22,655.28$ [5 marks]

[**Total 16 marks**]

4. (a) $200,000(1.06)^{\frac{1}{12}} - 1 = £973.51pm$
$20 \times 12 \times £973.51 = £233,642.43$

[2 marks]

(b) $Xs_{\overline{10}|}^{(12)} = 200,000$
$s_{\overline{10}|}^{(12)} = \frac{(1+i)^{10}-1}{12((1+i)^{\frac{1}{12}}-1)} = 15.80989$ where $i = 9\%$
$\Rightarrow X = £12,650.31pa = £1,054.19pm$
Or use $s_{\overline{120}|} = \frac{(1+j)^{120}-1}{j} = 189.719$ where $j = .007207323$

[3 marks]

(c) $s_{\overline{120}|} = \frac{(1+j)^{120}-1}{j} = 154.363$ where $j = 1.05^{\frac{1}{12}} - 1 = 0.004074124$
Outstanding loan $= 200,000 - 154.363 \times 1,054.1926 = £37,271.51$

[3 marks]

(d) Let level repayment be X where $37,271.51 = Xa_{\overline{15}|}^{(12)}$ at 7%
$a_{\overline{15}|}^{(12)} = 112.7587 \Rightarrow X = £330.542$

[2 marks]

(e) $200,000 = (1,054.19 + 973.51)a_{\overline{10}|}^{(12)} + 330.54v^{10}a_{\overline{15}|}^{(12)}$
$= 179,692 + 20,296 = 200,000$ where $i = 6.55\%$
So 6.55% is the yield

[4 marks]

[**Total 14 marks**]

5. Check $\frac{D(1-t_1)}{R} = 5.45\% > 5\%$ so no CGT

(a) $P = 1,000((0.75 \times 4)a_{\overline{30}|} + 110v^{30} = 63.0469 + 52.9119) = £115,958.81$
where $j = \sqrt{1.05} - 1$ [4 marks]

(b) $P = 1,000(3a_{\overline{40}|} + 110v^{40} = £75.6965 + £41.4578) = £117,154.37$

[2 marks]

(c) As soon as possible as the interest he is paying is greater than the market rate [2 marks]

[**Total 8 marks**]

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